AKASH SRINIVASAN

AI & Data Science Student — AI/ML & GenAI Enthusiast

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PROFESSIONAL SUMMARY

B.Tech Artificial Intelligence & Data Science undergraduate with hands-on experience in Machine Learning, Computer Vision, NLP, and web-based AI applications. Experienced in building AI/ML prototypes and web-based applications using Python, machine learning, computer vision, APIs, and modern web technologies. Freelance AI and software developer interested in industrial AI, digital twins, Generative AI, and LLM applications.

EDUCATION

Vel Tech High Tech Engineering College **2026 – Present**
B.Tech – Artificial Intelligence & Data Science **CGPA: 7.8/10**

Adarsh Matric Higher Secondary School **2023 – 2024**
Higher Secondary (Class XII) **80%**

Adarsh Matric Higher Secondary School **2021 – 2022**
SSLC (Class X) **75.2%**

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, HTML, CSS
AI / Machine Learning: Machine Learning, Deep Learning, NLP, Computer Vision
Generative AI: Generative AI, LLM Concepts, RAG, Prompt Engineering
Frameworks / Libraries: Scikit-learn, XGBoost, TensorFlow, Pandas, NumPy, Flask, FastAPI, Streamlit, ReactJS
Databases: MySQL, MongoDB, NoSQL
Tools & Platforms: Git, GitHub, VS Code, Google Colab

EXPERIENCE

Freelance AI & Software Developer **2025 – Present**

- Develop AI/ML and software prototypes based on real-world problem requirements.
- Build web-based applications, AI prediction systems, dashboards, and automation solutions using Python and modern development frameworks.
- Work across project planning, development, testing, debugging, and deployment.
- Translate functional requirements into practical, user-focused software solutions.

Computer Vision Intern – NSIC Technical Services Centre **Jun 2025 – Jul 2025**

- Worked on image detection, feature extraction, and real-time object recognition pipelines.
- Applied supervised machine learning concepts and CNN-based models through hands-on mini-projects.

PROJECTS

AutoTwin-QI – Digital Twin for Industrial Quality Inspection

- Developing an AI-assisted quality inspection platform to analyze industrial components and support defect assessment.
- Designing a modular FastAPI backend for inspection results, engineering compliance checks, quality-risk scoring, and reports.
- Planning a 3D digital-twin interface and what-if simulation to visualize defects and compare estimated process-risk scenarios.
- Integrating root-cause analysis and Corrective and Preventive Action (CAPA) workflows to support quality-engineer decisions.

Technologies: Python, FastAPI, React, YOLO, Computer Vision, Three.js, PostgreSQL

AI-Based Crop Recommendation System

- Developed an ML-based application that recommends suitable crops using soil, weather, and agricultural parameters.
- Implemented data preprocessing, model training, and real-time prediction through a Flask-based backend.

Technologies: Python, Pandas, NumPy, Scikit-learn, Flask

NLP Healthcare Symptom Checker

- Developed an NLP-based application to process user-provided symptoms and generate preliminary health-related information.
- Built an interactive application using Python and machine learning techniques.

Technologies: Python, NLP, Machine Learning, Streamlit/Flask

AI Attendance System & Student Management

- Built a face recognition-based attendance system with automated attendance tracking and student record man-

agement.
- Implemented automated attendance recording and report generation using computer vision techniques.
- Designed the system to simplify student attendance management and reduce manual record keeping.

Technologies: Python, OpenCV, Computer Vision

CERTIFICATIONS

- **Retrieval-Augmented Generation for Enhanced AI Outputs** – IBM SkillsBuild, 2026
- **Generative AI for Data Analytics** – Udemy, 4.5 hours, Aug 2026
- **Introduction to Machine Learning** – 2026
- **NLP with Python & NLTK** – 2026
- **Responsive Web Development with HTML5, CSS3, JavaScript** – Infosys Springboard, Sep 2026
- **Blockchain & Hyperledger Fabric: An Overview of Blockchain Technology** – Infosys Springboard, Aug 2026
- **Introduction to Internet of Things** – Cisco Networking Academy, Aug 2026
- **Smart Sensor Networks (Advanced)** – FutureSkills Prime, Aug 2026
- **R Programming for Beginners** – Simplilearn SkillUp, Aug 2026

HACKATHONS & ACHIEVEMENTS

- **Singularity Hackathon – AJ Institute of Engineering & Technology** *2026*
  Currently participating in the hackathon, collaborating with team members to develop an innovative technology-based solution.
- **Smart India Hackathon (SIH) 2026 – Internal Hackathon, Vel Tech High Tech Engineering College** – Won the internal hackathon by developing and presenting a solution for an SIH 2026 problem statement.
- **Bharatiya Antariksh Hackathon 2026 – ISRO** – Successfully submitted an idea addressing a real-world space-related challenge.
- **Zyren Hackathon 2026 – SRM Ramapuram** – Participated in a competitive hackathon and developed a technology-driven solution.
- **Smart India Hackathon (SIH) 2025** – Selected as a participant and presented an idea in the Internal SIH round at Vel Tech.
- **MECHGUST'25** – Participated at Chennai Institute of Technology.
- **QUIZOCALYPSE** – Participated at Rajalakshmi Engineering College.
- **Astrava Hackathon 2026** – Participated at Dr. Ambedkar Institute of Technology.

CORE COMPETENCIES

Analytical Thinking — Problem Solving — Team Collaboration — Communication — Leadership